Targeted Oxidative Weakening of Woody Invasive Root Crowns Using Cold Atmospheric Plasma: The Pulsed Micro-Plasma Root Interface (PMRI)

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Abstract

Cold atmospheric plasma (CAP) is an emerging precision technology in agriculture and environmental biotechnology, capable of generating complex mixtures of reactive oxygen and nitrogen species (ROS/RNS) at near-ambient temperatures. These chemically active plasmas have demonstrated effectiveness in seed treatment, pathogen suppression, and lignocellulosic biomass modification.

However, the targeted application of CAP to the living woody root crowns of established perennial invasive plants has not been explored.

Here we introduce a conceptual Pulsed Micro-Plasma Root Interface (PMRI) — a low-cost handheld CAP wand designed to deliver controlled bursts of ROS/RNS directly to exposed root crowns of brambles (*Rubus* spp.) following mechanical cutting. The plasma plume produces a localized flux of oxidative species including hydroxyl radicals, ozone, hydrogen peroxide, and nitrogen oxides, which interact with lignified vascular tissues at the crown surface. By selectively fragmenting lignin β-O-4 aryl ether linkages within these tissues, PMRI is hypothesized to induce structural weakening, carbohydrate leakage, oxidative membrane damage, and subsequent microbial colonization, ultimately leading to progressive crown necrosis.

Unlike thermal or chemical weed control methods, the proposed approach operates at low temperatures and leaves no persistent chemical residues, minimizing risks of soil disturbance, collateral vegetation damage, or contamination of terrace-edge environments. The method is particularly suited to wet-season conditions, where elevated humidity enhances hydroxyl radical formation and facilitates penetration of reactive species into exposed crown tissues.

PMRI integrates with simple mechanical cutting and crown marking protocols to create an erosion-safe control strategy for invasive *Rubus* populations on steep slopes, terraces, and riparian margins where conventional digging or herbicide application is problematic. We describe the conceptual framework of the approach, a prototype device build ($<$ 150 USD), operational treatment protocols, and a proposed small-plot randomized field trial evaluating regrowth suppression, crown viability, lignin structural modification, and soil microbial responses. Potential pathways for scaling the technology through semi-autonomous or drone-assisted treatment systems are also discussed.

To our knowledge, this represents the first proposed application of CAP for targeted weakening and control of established woody perennial invasives via treatment of exposed root crowns.

Keywords: cold atmospheric plasma, plasma agriculture, invasive plant management, Rubus spp., lignin oxidation, non-thermal plasma treatment, ecological restoration

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1 Introduction

1.1 Invasive *Rubus* and Challenges in Crown Control

Himalayan blackberry and related *Rubus* species form dense, thorny thickets that dominate terrace edges, riparian zones, forest margins, and disturbed land across temperate regions worldwide [15, 16]. Introduced populations of *Rubus armeniacus* and closely related taxa are particularly invasive in North America, Europe, Australia, and parts of South America. Their vigorous vegetative growth and extensive crown-based regeneration allow them to rapidly outcompete native vegetation and form persistent monocultures.

These thickets alter light regimes, reduce biodiversity, obstruct access to agricultural land, and accelerate bank instability along slopes and waterways. Because regrowth originates from the perennial crown and root system, established infestations can persist for many years even after repeated mechanical disturbance.

Control of established *Rubus* stands therefore remains challenging. Following cutting or grazing, dormant buds located on the crown rapidly initiate new cane growth using carbohydrates stored in lignified vascular tissues. Effective eradication typically requires repeated mechanical cutting, excavation of root crowns, or application of systemic herbicides such as glyphosate or triclopyr.

Each of these approaches carries significant limitations, particularly on steep or terraced land. Mechanical removal can destabilize soil and increase erosion risk, while repeated cutting is labour-intensive and often only suppresses regrowth temporarily. Chemical control methods raise concerns regarding environmental contamination, non-target plant damage, and regulatory restrictions in sensitive riparian or agricultural zones.

1.2 Cold Atmospheric Plasma in Agricultural Biotechnology

Cold atmospheric plasma (CAP) has recently emerged as a versatile tool in agricultural biotechnology. CAP refers to partially ionized gases generated at atmospheric pressure that produce chemically reactive environments containing reactive oxygen and nitrogen species (ROS/RNS) while maintaining near-ambient gas temperatures [20, 13].

Typical CAP-generated species include hydroxyl radicals ($\bullet$OH), ozone (O_3), hydrogen peroxide (H_2O_2), nitric oxide (NO), and nitrogen dioxide (NO_2). These species are highly oxidative and capable of inducing chemical modification of organic materials, including disruption of lignocellulosic structures such as lignin and cellulose.

Extensive research has demonstrated the ability of non-thermal plasma to modify lignin and other structural polymers in plant biomass. In the context of biofuel and biorefinery processing, plasma treatments have been shown to cleave lignin β-O-4 aryl ether bonds, increase surface porosity, and enhance enzymatic digestibility of lignocellulosic materials [1, 2, 3, 4].

In parallel, agricultural applications of CAP have expanded rapidly in recent years. Documented uses include seed germination enhancement, surface sterilization of plant pathogens, stimulation of plant growth responses, and inactivation of weed seeds prior to germination [5, 6, 7, 8]. These effects arise from oxidative interactions between plasma-generated species and biological tissues, including disruption of cell membranes, modification of surface chemistry, and induction of localized stress responses.

1.3 Research Gap and Conceptual Approach

Despite these advances, existing CAP applications in agriculture have focused almost exclusively on seeds, pathogens, or detached plant material. To date, no published studies have explored the targeted application of CAP to the living woody root crowns of established perennial weeds.

The crown represents a critical regenerative structure in many invasive woody plants, including *Rubus* species. This structure contains dormant buds and vascular tissues that enable

rapid regrowth after disturbance. Damage to these tissues—particularly the outer lignified layers responsible for carbohydrate transport and bud initiation—has the potential to significantly weaken the plant's regenerative capacity.

In this work we propose the Pulsed Micro-Plasma Root Interface (PMRI) as a precision follow-up treatment to mechanical cutting. After canes are cut low and crowns are marked, a handheld CAP wand delivers pulsed plasma directly to the exposed or near-surface crown.

The localized plasma plume generates a concentrated flux of reactive species that interact with lignified crown tissues, promoting oxidative degradation of structural polymers and inducing localized physiological stress. Wet rain-season soil acts as a natural humidity coupler, enhancing ROS formation—particularly hydroxyl radicals—and facilitating penetration of reactive species into surface crown tissues. The method is particularly suited for terrace edges where digging risks bank collapse and soil disturbance must be minimized.

The proposed approach offers several conceptual advantages over conventional methods. Because CAP operates at near-room temperature, it avoids the thermal damage associated with flame weeding while leaving no persistent chemical residues in soil or water. Treatment can be applied selectively to individual crowns following mechanical cutting, minimizing disturbance to surrounding vegetation and soil structure. Moreover, oxidative weakening of crown tissues may promote microbial colonization and gradual necrosis of the crown, reducing the plant's ability to regenerate new canes.

1.4 Scope of the Present Work

This paper outlines the scientific basis for the Pulsed Micro-Plasma Root Interface concept, describes a low-cost prototype device and treatment protocol, and proposes a small-plot experimental framework to evaluate its efficacy in field conditions. Expected outcomes, safety considerations, and potential future developments—including semi-autonomous or drone-assisted treatment systems—are also discussed.

If validated through field trials, PMRI could represent a novel non-thermal approach for targeted weakening and control of woody perennial invasive plants.

Contributions of this work The present study introduces and formalizes the Pulsed Micro-Plasma Root Interface (PMRI) concept as a potential new approach for invasive woody plant management. The main contributions of this work are:

- Introduction of a conceptual plasma-assisted method for targeted weakening of woody invasive plant crowns following mechanical cutting.
- Design description of a low-cost handheld cold atmospheric plasma device suitable for field deployment.
- Development of a mechanistic framework linking plasma physics, reactive species chemistry, lignin oxidation, and crown physiology.
- Proposal of an experimental framework for validating plasma-assisted crown weakening through controlled small-plot field trials.
- Preliminary engineering estimates of energy delivery and reactive species exposure relevant to plasma–plant interactions.

Broader significance for invasive plant management Woody invasive plants represent a persistent global challenge for ecological restoration and land management. Species such as *Rubus* spp. regenerate rapidly from crown tissues after mechanical disturbance, often requiring repeated cutting, herbicide application, or soil excavation for effective control. These approaches

can be labour intensive, environmentally disruptive, or restricted in sensitive landscapes such as riparian corridors, terraces, and steep slopes. Consequently, there is growing interest in alternative control strategies that minimize soil disturbance, avoid persistent chemical inputs, and enable precise treatment of individual plants. In this context, emerging technologies capable of delivering localized biochemical stress to plant tissues may offer new possibilities for targeted invasive plant management. The PMRI concept explored in this study represents one potential example of such an approach.

1.5 Conceptual Positioning of the PMRI Approach

The present work is intended as a conceptual engineering study rather than a report of experimentally validated results. Its primary objective is to articulate and formalize the Pulsed Micro-Plasma Root Interface (PMRI) as a physically plausible approach for targeted weakening of woody invasive plant crowns using cold atmospheric plasma.

The paper therefore synthesizes existing knowledge from atmospheric plasma physics, reactive species chemistry, lignocellulosic material modification, and plant physiology to construct a coherent mechanistic framework linking plasma exposure to suppression of crown regrowth. Prototype device architecture, treatment protocols, and experimental validation strategies are presented to enable future empirical investigation.

Such conceptual technology studies are commonly used in emerging interdisciplinary fields to identify unexplored applications of established physical processes and to guide subsequent experimental research programs.

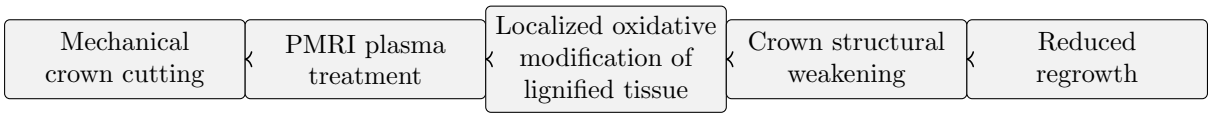


Figure 1: Conceptual system architecture of the PMRI treatment pipeline. Mechanical cutting exposes the *Rubus* crown surface, enabling targeted plasma application. Reactive oxygen and nitrogen species generated by the atmospheric plasma induce localized oxidative modification of lignified tissues, leading to structural weakening of the crown and reduced regenerative capacity.

2 Materials and Methods

2.1 Prototype Construction

The prototype described here represents a reference implementation of the Pulsed Micro-Plasma Root Interface (PMRI) concept intended to demonstrate engineering feasibility using widely available components.

A handheld cold atmospheric plasma (CAP) wand was constructed using low-cost commercially available components in a dielectric barrier discharge (DBD) configuration. The device was designed to generate a non-thermal plasma plume at atmospheric pressure while maintaining minimal thermal transfer to plant tissue.

- **High-voltage driver:** 12–24 V zero-voltage-switching (ZVS) driver coupled to a compact flyback transformer or DC–DC plasma module, producing a pulsed output in the range of 8–20 kV peak-to-peak at 10–40 kHz.
- **Dielectric barrier:** quartz capillary tube (inner diameter 2–4 mm, length 10–20 cm) serving as the dielectric barrier and plasma channel.
- **Electrodes:** central stainless-steel needle electrode (diameter 1–2 mm) and external brass or copper ring electrode positioned near the capillary outlet to stabilize the plasma plume.

- **Power supply:** 12 V lithium polymer battery pack (3–5 Ah) carried in a small backpack module for field mobility.
- **Support structure:** insulated fiberglass pole (1.5–2 m length) allowing safe operation while maintaining distance from dense bramble thickets.
- **Pulse control (optional):** Arduino Nano or equivalent microcontroller enabling duty-cycle modulation and burst operation (1–10 kHz gating).

Under typical operation the device consumes 5–20 W of electrical power. Surface temperatures at the treatment point remain below approximately 40 °C, ensuring that plant damage is dominated by reactive chemical species rather than thermal effects. The plasma plume length is typically 1–3 cm under ambient air operation.

The device architecture follows established designs for portable cold atmospheric plasma (CAP) sources used in medical sterilization and agricultural plasma treatment systems [6, 13, 9].

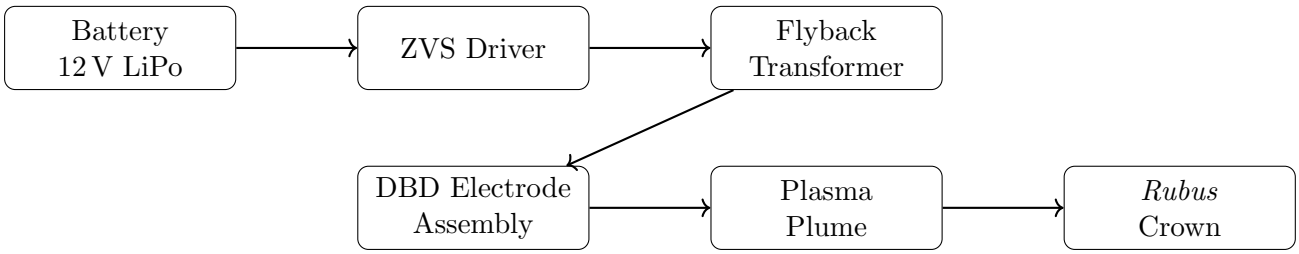
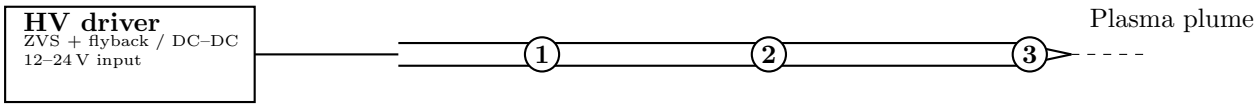


Figure 2: Conceptual architecture of the Pulsed Micro-Plasma Root Interface (PMRI) device and treatment pathway. Electrical power from a portable battery drives a high-voltage plasma discharge that generates a localized plume of reactive species applied to the exposed crown surface.



ID	Component description
1	**Quartz dielectric tube** — capillary / DBD barrier (10–20 cm), providing electrical insulation and stabilizing the discharge channel.
2	**Central HV electrode** — needle electrode (1–2 mm) positioned inside the dielectric tube to drive microdischarge formation.
3	**Plasma generation tip** — electrode geometry and outlet region producing the atmospheric plume directed toward the exposed crown surface.

Figure 3: Conceptual prototype of the PMRI plasma wand showing the quartz dielectric tube, high-voltage electrode, and atmospheric plasma plume generated at the tip. Numbered markers correspond to component descriptions in the table.

2.2 Treatment Protocol (Conceptual Field Trial)

Field treatment targets the exposed crown region of *Rubus* spp. following mechanical cutting.

1. Bramble canes are cut to 2–3 cm above the soil surface during wet-season conditions when soil moisture is high.

2. Visible crowns are marked using survey flags to maintain consistent spatial coverage (typical spacing 30–50 cm).
3. The crown surface is lightly misted with water to increase local humidity and promote formation of hydroxyl radicals during plasma exposure.
4. The plasma wand tip is positioned 3–10 cm above the exposed crown.
5. Plasma is applied in three cycles of 10 s exposure followed by 5 s pause (total treatment time approximately 45 s per crown).
6. A follow-up treatment is performed after 3–4 weeks if weak regrowth is observed.
7. Treated areas are immediately seeded with competitive grasses or mulched to suppress secondary weed colonization and stabilize soil on terrace edges.

Preliminary efficacy can be assessed using simple field indicators including delayed bud emergence, reduced cane diameter, increased crown desiccation, and reduced shoot density relative to cut-only controls.

2.3 Experimental Design Outline

A small-plot randomized block experiment is proposed to evaluate the efficacy of PMRI treatment.

- **Treatments:**
  1. PMRI treatment following mechanical cutting
  2. Mechanical cutting only (control)
  3. Propane flame weeding (thermal comparison)
- **Sample size:** $n = 30$ individual crowns per treatment group.
- **Plot layout:** treatments randomly assigned within terrace-edge blocks to minimize spatial bias related to soil moisture, slope position, and crown age.
- **Observation period:** 90 days following initial treatment.
- **Measured metrics:**
  - Regrowth biomass (dry weight at 30, 60, and 90 days)
  - Crown viability using tetrazolium chloride staining
  - Lignin structural changes assessed via FTIR spectroscopy or Klason lignin analysis
  - Soil microbial community shifts evaluated using 16S rRNA amplicon sequencing

3 Design Requirements and Operational Envelope

For practical field use, the Pulsed Micro-Plasma Root Interface (PMRI) system must operate within a set of engineering and environmental constraints. Defining these operational requirements helps clarify the feasibility of the concept and guides future prototype development.

3.1 Operational Requirements

The PMRI system is intended for use in dense bramble thickets and along terrace edges where conventional invasive plant control methods are difficult to apply. Key operational requirements include:

- Portable handheld operation by a single operator
- Low electrical power consumption to allow battery-powered field deployment
- Non-thermal plasma generation to avoid ignition of dry vegetation
- Minimal disturbance to surrounding soil and vegetation
- Targeted treatment of individual crowns following mechanical cutting

3.2 Representative Operating Parameters

Table 1 summarizes representative parameter ranges consistent with handheld dielectric barrier discharge plasma devices.

Table 1: Representative operating parameters for the PMRI concept.

Parameter	Typical Range	Design Rationale
Voltage amplitude	8–20 kV	Required for stable atmospheric plasma discharge
Operating frequency	10–40 kHz	Typical range for compact DBD drivers
Electrical power	5–20 W	Compatible with portable battery operation
Plasma plume length	1–3 cm	Allows localized treatment of crown surface
Treatment distance	3–10 cm	Maintains operator safety and plume stability
Exposure duration	∼45 s per crown	Sufficient reactive species dose for oxidative interaction

3.3 Environmental Operating Conditions

The PMRI concept is particularly suited to humid outdoor environments where atmospheric moisture enhances formation of hydroxyl radicals and other reactive oxygen species. Typical operating conditions include:

- Ambient air plasma operation (no external process gas required)
- Moderate to high relative humidity (¿50%)
- Moist crown surfaces following rainfall or irrigation
- Outdoor operation allowing rapid dispersion of reactive gases

These environmental conditions are commonly encountered during wet-season invasive plant management, particularly on terrace edges and riparian margins.

Representative Operating Envelope of the PMRI Concept

Operating voltage: 8–20 kV peak-to-peak
Operating frequency: 10–40 kHz dielectric barrier discharge
Electrical power: 5–20 W handheld plasma source
Treatment duration: ~ 45 s per crown
Estimated energy delivered: ~ 450 J per treatment cycle
Estimated treatment energy density: $\sim 10^2$ J cm^{-2}

4 Expected Outcomes and Mechanism

The conceptual biological pathway linking plasma exposure to suppression of crown regrowth is illustrated in Figure 4.

Cold atmospheric plasma produces a chemically reactive environment containing reactive oxygen and nitrogen species (ROS/RNS), charged particles, and ultraviolet photons. These agents interact with biological tissues primarily through oxidative reactions that modify structural polymers, membrane lipids, and intracellular biomolecules.

Plasma-generated ROS/RNS oxidatively damage lignified vascular tissues, including cleavage of lignin β-O-4 aryl ether linkages, producing microstructural weakening within the outer 1–5 mm of the woody crown [1, 3]. Many *Rubus* species regenerate from shallow cambial and vascular tissues located near the outer crown surface. Disruption of these tissues can substantially impair the crown's ability to mobilize stored carbohydrates and initiate vigorous shoot regrowth, even when deeper structural tissues remain intact.

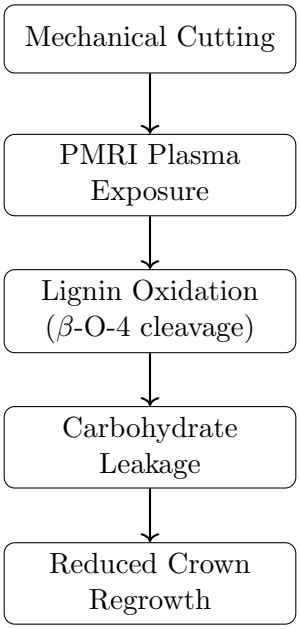


Figure 4: Conceptual biological pathway of PMRI treatment illustrating the proposed sequence of processes linking plasma exposure to suppression of *Rubus* crown regrowth. Mechanical cutting exposes crown tissues, plasma-generated reactive oxygen and nitrogen species induce oxidative modification of lignified structures, and subsequent metabolic disruption leads to reduced regenerative capacity.

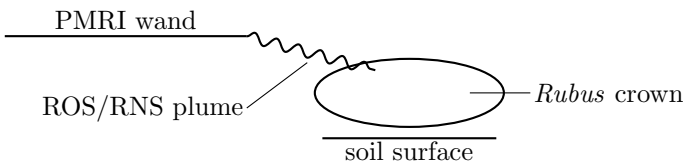


Figure 5: Conceptual depiction of the plasma plume interacting with the exposed *Rubus* crown surface during PMRI treatment. Reactive oxygen and nitrogen species generated in the atmospheric plasma contact the crown tissues and induce localized oxidative modification.

4.1 Reactive Species Involved

Atmospheric plasma operating in ambient air generates a mixture of short- and long-lived reactive species that contribute to plant tissue damage. Key species include:

- Hydroxyl radicals (•OH) – extremely strong oxidants capable of attacking lignin and membrane lipids
- Ozone (O_3) – oxidizes phenolic structures in lignified tissues
- Hydrogen peroxide (H_2O_2) – diffuses into tissue and contributes to oxidative stress
- Nitric oxide (NO) and nitrogen dioxide (NO_2) – reactive nitrogen species that interfere with cellular metabolism
- Ultraviolet photons – contribute to DNA and protein damage

Table 2: Major reactive species generated in cold atmospheric plasma and their biological effects on plant tissues.

Species	Chemical Type	Primary Biological Effect
•OH	Hydroxyl radical	Strong oxidation of lignin and membrane lipids
O_3	Ozone	Oxidative modification of phenolic polymers
H_2O_2	Hydrogen peroxide	Diffusive oxidative stress within tissues
NO / NO_2	Reactive nitrogen species	Interference with cellular signaling and metabolism
UV photons	Electromagnetic radiation	DNA and protein damage

4.2 Proposed Crown Weakening Mechanism

The expected weakening of *Rubus* crowns following plasma exposure likely occurs through a sequence of interacting processes:

- Oxidative degradation of lignin and structural polymers in the outer crown tissues
- Increased permeability of cell walls and membranes
- Leakage of stored carbohydrates and metabolites
- Induction of localized oxidative stress within crown cells
- Disruption of cambial and vascular tissues responsible for regrowth
- Increased susceptibility to microbial colonization and decay

Stored carbohydrates leak, oxidative stress disrupts cell membranes, and soil microbes invade the compromised tissue. Regrowth emerges weaker and slower, allowing repeated low-effort mowing or follow-up plasma to exhaust the crown completely.

4.3 Role of Crown Physiology

The regenerative capacity of *Rubus* species is concentrated near the crown surface, where dormant buds and vascular tissues coordinate shoot emergence following disturbance. These tissues are responsible for:

- Mobilization of stored carbohydrate reserves
- Activation of dormant buds after cutting
- Transport of water and nutrients to emerging canes

Because these structures occur within the outer crown layers, localized oxidative damage within the first few millimeters of tissue may significantly impair the plant's regenerative capacity even when deeper structural roots remain intact.

4.4 Environmental Enhancement Effects

Environmental conditions can influence plasma treatment efficacy.

- High humidity increases formation of hydroxyl radicals
- Wet soils enhance electrical coupling and plasma stability
- Moist crown surfaces facilitate diffusion of reactive species

In wet conditions, increased •OH radical formation enhances efficacy. Penetration is surface-limited on thick crowns, so PMRI is best framed as a powerful *weakening* step rather than instant eradication.

Table 3: Conceptual stages of crown weakening following PMRI treatment.

Stage	Primary Process	Expected Outcome
Immediate	ROS/RNS oxidation of lignified tissues	Surface tissue damage and increased porosity in outer crown layers
Short-term (days)	Membrane disruption and metabolic stress	Reduced vitality of crown tissues and leakage of stored carbohydrates
Intermediate (weeks)	Microbial colonization and tissue decay	Progressive weakening of vascular and cambial structures
Long-term (weeks–months)	Energy reserve depletion	Reduced regrowth vigor and eventual exhaustion of the crown

4.5 Order-of-Magnitude Feasibility Estimate

A simple energetic estimate provides an approximate scale for plasma exposure during PMRI treatment. Assuming a nominal plasma power of 10 W and an exposure duration of 45 s per crown (Section 2.2), the electrical energy delivered to the discharge is:

$$E = P \times t$$

$$E \approx 10\,\mathrm{W} \times 45\,\mathrm{s} = 450\,\mathrm{J}$$

Handheld atmospheric plasma sources operating at similar power levels are known to generate reactive oxygen and nitrogen species (ROS/RNS) at concentrations sufficient to induce measurable oxidative modification of lignocellulosic materials [1, 3].

Given a plasma plume diameter of approximately 2 cm, the treated surface area is roughly:

$$A \approx \pi r^2 \approx 3.1\,\mathrm{cm}^2$$

This suggests that several hundred joules of electrical energy may be delivered to the plasma discharge during a typical treatment cycle. In atmospheric plasma systems, only a fraction of the electrical input energy is converted into chemically active species such as reactive oxygen and nitrogen species (ROS/RNS), with the remainder dissipated through ionization losses, gas heating, and radiation. Even with modest conversion efficiencies, however, localized generation of strong oxidants such as hydroxyl radicals and ozone may produce measurable oxidative modification of lignified crown tissues. Even partial oxidative modification of lignified vascular tissues within the outer millimeters of the crown may plausibly impair carbohydrate transport and reduce regenerative capacity. These calculations are intended only as order-of- magnitude estimates; experimental measurements will be required to determine the true plasma coupling efficiency and biological response.

Expressed as an approximate surface exposure, the delivered electrical energy corresponds to an energy density on the order of

$$E_A \approx \frac{E}{A} \approx \frac{450\ \mathrm{J}}{3.1\ \mathrm{cm}^2} \approx 1.4 \times 10^2\ \mathrm{J\ cm}^{-2}$$

This value represents only a coarse order-of-magnitude estimate because the true fraction of electrical energy converted into chemically active species and delivered to plant tissue depends on plasma coupling efficiency, plume geometry, and environmental conditions.

5 Operator Safety and Human Exposure

Cold atmospheric plasma (CAP) devices are increasingly used in medical and biomedical applications, including treatment of chronic wounds, skin ulcers, dental infections, and mucosal surfaces. Clinical and laboratory studies have demonstrated that non-thermal atmospheric plasma can be applied to human tissue with minimal adverse effects when exposure parameters are appropriately controlled [11, 12]. These medical plasma systems operate by generating reactive oxygen and nitrogen species (ROS/RNS) at near-room temperature, allowing antimicrobial and tissue-modifying effects without thermal injury.

Short exposures of CAP similar to those proposed in the PMRI treatment protocol have been widely used in medical plasma devices for wound disinfection and tissue stimulation. These studies generally report low levels of cytotoxicity and no significant long-term adverse effects when exposure durations remain within established operational ranges. Because the PMRI system is designed for intermittent short bursts of plasma applied at a distance of several centimeters from the operator, direct human exposure is expected to be substantially lower than that encountered in many clinical plasma treatments.

Potential operator exposure pathways include:

- incidental contact with the plasma plume
- inhalation of trace ozone or nitrogen oxides
- exposure to ultraviolet radiation emitted by the discharge
- electrical hazards associated with high-voltage components

Under typical open-air operating conditions, reactive species generated by atmospheric plasma disperse rapidly and decay on millisecond to second timescales. Ozone concentrations produced by small portable plasma devices are generally low and dissipate quickly in outdoor environments. Similarly, ultraviolet emissions from dielectric barrier discharge plasma sources are weak compared with conventional UV lamps and decrease rapidly with distance from the source.

The PMRI device design further reduces operator exposure through several engineering features:

- operation at low electrical power (typically 5–20 W)
- short plasma plume length (approximately 1–3 cm)
- use of an insulated fiberglass pole allowing operation at arm's length
- intermittent duty cycle limiting continuous plasma generation

Basic personal protective equipment (PPE) is recommended during operation:

- safety goggles to protect against incidental ultraviolet emission
- insulated gloves when handling the device
- avoidance of prolonged inhalation of plasma exhaust gases

Table 4: Potential exposure pathways and recommended mitigation measures during PMRI operation.

Exposure Pathway	Potential Risk	Mitigation Measure
Direct plasma contact	Mild skin irritation or localized oxidative stress	Maintain 3–10 cm operating distance from crown surface
Ozone and reactive gases	Temporary respiratory irritation at high concentrations	Outdoor operation and natural atmospheric dispersion
Ultraviolet emission	Eye irritation with prolonged direct viewing	Use protective safety goggles
Electrical hazards	Shock from high-voltage components	Insulated pole design and enclosed power electronics

Overall, when operated outdoors with basic protective equipment, the PMRI system presents a relatively low risk profile compared with many conventional weed-control methods such as flame weeding, herbicide spraying, or mechanical excavation. Nevertheless, as with any electrical field device, standard safety precautions and responsible operation are essential. Future prototype development may incorporate additional safeguards such as interlock switches, grounded housings, and automatic duty-cycle limits to further enhance operational safety.

6 Limitations

The present work is conceptual in nature and is intended to establish prior art and outline a physically plausible approach rather than to report experimentally validated results. No empirical field trials, laboratory measurements, or controlled plasma exposure experiments were conducted as part of this study. Consequently, the mechanisms and expected outcomes described herein are derived from synthesis of existing literature in plasma chemistry, lignocellulosic material modification, and invasive plant physiology rather than from direct experimental observation.

Several key uncertainties therefore remain.

- **Plasma penetration depth:** Atmospheric plasma interactions are generally limited to surface layers of biological material. The extent to which reactive species can penetrate and damage lignified crown tissues in living *Rubus* plants remains to be experimentally quantified.
- **Dose-response relationships:** The optimal plasma exposure duration, duty cycle, and distance from the crown required to achieve meaningful physiological damage are currently unknown. Excessively short exposures may produce negligible effects, while longer exposures may offer diminishing returns.
- **Plant physiological response:** Perennial plants possess strong regenerative capabilities. It is possible that partial oxidative damage to crown tissues could trigger compensatory growth responses rather than long-term suppression under some conditions.
- **Environmental variability:** Soil moisture, humidity, crown size, plant age, and microbial community composition may all influence treatment effectiveness. The extent of these environmental interactions remains to be studied.
- **Operational scalability:** While the PMRI device is designed as a handheld system, practical treatment rates, operator fatigue, and large-scale deployment logistics require evaluation under realistic field conditions.

Future research could address these uncertainties through controlled laboratory and field experiments. Initial validation studies might include bench-scale plasma exposure of excised *Rubus* crown tissue to measure structural changes in lignin and cell walls using spectroscopic techniques such as FTIR or Raman spectroscopy. Subsequent greenhouse or small-plot trials could quantify regrowth suppression following plasma treatment compared with conventional cutting controls. Measurements of crown viability, carbohydrate depletion, and microbial colonization would further clarify the biological pathways involved.

Field-scale trials on terrace edges or riparian margins would ultimately be required to evaluate the practical effectiveness of the PMRI approach under real environmental conditions. Such experiments could also assess treatment efficiency, energy consumption, and operator safety during extended use.

Despite these limitations, the conceptual framework presented here identifies a potentially unexplored intersection between atmospheric plasma technology and invasive plant management. Establishing this prior-art concept provides a foundation for future experimental investigation and technological development.

Reactive species footprint validation. A particularly important early validation step for the PMRI concept is experimental confirmation that the plasma plume delivers a measurable and spatially localized flux of reactive oxygen and nitrogen species (ROS/RNS) at crown-relevant treatment distances. Because the proposed biological mechanism depends on oxidative exposure at the crown surface, demonstration of a repeatable oxidant footprint represents a critical intermediate validation between electrical plasma generation and biological response. A practical approach is the use of colorimetric indicator substrates placed at the treatment surface, such as potassium iodide–starch paper for oxidant detection or Griess reagent strips as proxies for reactive nitrogen species. Under controlled electrode-to-surface distances and duty cycles representative of field operation, plasma exposure should produce a localized colorimetric response that can be imaged and quantified in terms of footprint area and relative intensity. Although such measurements do not directly measure lignin modification or physiological damage, they provide a simple and inexpensive method to verify that the plasma source delivers reproducible

oxidative exposure to the target surface. Confirmation of this oxidative footprint would constitute an important first experimental step in validating the physical feasibility of plasma-assisted crown weakening.

A more detailed description of this experimental approach is outlined in Section 6.4.

6.1 Experimental Measurements Required for Concept Validation

Several engineering measurements will be necessary to experimentally validate the PMRI concept and quantify plasma–plant interactions under field conditions. The present study does not include such measurements, but they represent important targets for future experimental work.

Key measurements include:

- **Plasma plume temperature.** Measurement of gas temperature within the plasma plume using thermocouples or optical emission spectroscopy would confirm that the discharge remains non-thermal and does not introduce significant heat damage to crown tissues.
- **Reactive species flux.** Quantification of reactive oxygen and nitrogen species (ROS/RNS) concentrations near the crown surface using optical emission spectroscopy, chemical probes, or ozone sensors would provide direct evidence of oxidative exposure during treatment.
- **Electrical waveform characterization.** Oscilloscope measurements of voltage and current waveforms during plasma operation would allow calculation of true discharge power and energy coupling efficiency.
- **Electrical power measurement.** Direct measurement of electrical power delivered to the discharge using high-voltage probes and power analyzers would refine the order-of-magnitude energy estimates presented in Section 4.5.
- **Prototype device documentation.** Construction and documentation of a working prototype—including photographs of the device architecture and plasma plume—would provide empirical validation of the engineering feasibility described in Section 2.

Such measurements would enable quantitative characterization of plasma operating parameters, reactive species exposure, and energy coupling efficiency, thereby providing a stronger experimental foundation for evaluating the biological effects of PMRI treatment.

• Prototype device documentation. Construction and documentation of a working prototype—including photographs of the assembled device and the visible plasma plume—would provide direct visual confirmation of the engineering feasibility described in Section 2 and allow comparison of plume geometry, stability, and treatment distance under field-relevant conditions.

6.2 Engineering Validation Requirements

Because the PMRI concept has not yet been experimentally implemented, several engineering measurements and system-level analyses will be required to validate the device architecture and plasma–plant interaction model proposed in this study.

The following validation domains represent key experimental and engineering tasks for future prototype development.

1. **Electrical Characterization**

   Accurate characterization of the plasma discharge is required to determine true operating conditions.

   - Voltage waveform measurement using high-voltage probes

- Current waveform measurement
- True discharge power calculation
- Energy delivered per pulse
- Plasma stability during operation

Typical diagnostics include oscilloscope measurements and Lissajous figure analysis commonly used for dielectric barrier discharge (DBD) power estimation.

2. **Plasma–Tissue Coupling Mechanisms**

   Experimental investigation of the interface between the plasma plume and living crown tissue will be required to validate the proposed biological mechanism.

   Key processes include:

   - Diffusion of ROS/RNS into moist crown tissues
   - Surface oxidation of lignified structures
   - Moisture-assisted plasma chemistry
   - Microdischarge contact effects at the tissue interface

   These processes determine how reactive species generated in the plasma plume interact with crown tissues.

3. **Environmental Plasma Behaviour**

   Outdoor atmospheric plasma operation may differ significantly from controlled laboratory discharges.

   Relevant environmental factors include:

   - Wind effects on plume stability
   - Ambient humidity
   - Crown moisture and soil conductivity
   - Outdoor dispersion of reactive species

   Evaluation under realistic field conditions will therefore be required.

4. **Treatment Energy Density**

   The order-of-magnitude estimate presented in Section 4.5 suggests that approximately 450 J of electrical energy may be delivered during a typical 45 s treatment cycle.

   Assuming an effective treatment area of approximately $3\,\mathrm{cm}^2$, this corresponds to a rough treatment energy density on the order of

   $$\sim 10^2 \text{ J cm}^{-2}$$

   at the crown surface.

   Although simplified, expressing the treatment in terms of energy density provides a useful framework for comparing plasma exposure levels with other oxidative biomass modification processes.

5. **Device Scalability Considerations**

   Future development may explore alternative deployment architectures:

- Handheld systems for targeted crown treatment
- Backpack-mounted plasma power units
- Robotic ground platforms
- Drone-assisted plasma treatment systems

Engineering analysis of energy consumption, system mass, and treatment rate will determine practical scalability.

6. **System Architecture and Operational Throughput**

   The PMRI device can be interpreted as a system consisting of several interacting subsystems:

   - Portable power subsystem
   - Plasma generation subsystem
   - Optional control electronics
   - Operator interface

   Operational modeling should evaluate:

   - Crowns treated per hour
   - Energy consumption per crown
   - Battery runtime
   - Area treated per day

   Preliminary estimates in Appendix F suggest that several hundred crowns per day may be treated under typical conditions, though this remains to be experimentally verified.

7. **Potential Failure Modes**

   Several operational risks should be evaluated during prototype testing:

   - Plasma instability under heavy rain or strong wind
   - Insufficient reactive species penetration in thick crowns
   - Electrode fouling due to soil or plant debris
   - Electrical coupling through wet soil
   - Variability introduced by operator technique

   Identification and mitigation of these failure modes will be important for reliable field deployment.

Electrical parameter tuning and plume control. The geometry and electrical characteristics of atmospheric-pressure plasma jets are highly tunable, and several parameters can significantly modify the structure, length, and reactive species composition of the plasma plume. Key controllable variables include the applied voltage amplitude, operating frequency, waveform shape, and pulse duty cycle, all of which influence electron density, discharge power, and microdischarge behavior within dielectric barrier discharge systems. In addition, the electrical capacitance of the discharge region—determined by electrode spacing, dielectric barrier thickness, and dielectric material properties—affects the effective electric field and the number of microdischarges generated during each cycle. Modifying the dielectric tube material (e.g., quartz, alumina, or glass), its thickness, or the electrode geometry can therefore alter plasma

stability and energy coupling efficiency. Electrode configuration (needle, ring, or coaxial geometries), electrode spacing, and the presence of auxiliary or floating electrodes can also influence plume length and reactive species production. Environmental and operational factors further contribute to plume characteristics, including gas composition, humidity, gas flow rate, and the distance between the plasma source and the treated surface. Collectively, these parameters provide multiple engineering pathways for adjusting plume length, spatial confinement, and oxidative species flux at the treatment surface. Systematic exploration of these variables will be important during prototype optimization to tailor the PMRI plasma plume for efficient crown-surface oxidation while maintaining stable non-thermal operation.

In principle, these parameters could be optimized to generate either a narrowly confined treatment plume for precise crown targeting or a broader reactive footprint depending on the operational requirements of field deployment.

The ignition and stability of the discharge are also governed by the well-known Paschen scaling relationship, which links breakdown voltage to the product of gas pressure and electrode separation. In atmospheric-pressure systems this can be expressed approximately as

$$V_b = \frac{Bpd}{\ln(Apd) - \ln[\ln(1 + 1/\gamma_{se})]},$$

where p is gas pressure, d is electrode gap distance, and A, B, and γ_{se} are gas-dependent constants. Consequently, adjustments to electrode spacing, dielectric thickness, or local gas composition can shift discharge thresholds and influence plume formation, stability, and energy coupling efficiency in the PMRI device.

Collisional regime considerations. At atmospheric pressure the plasma operates in a strongly collisional regime where electron transport is limited by frequent collisions with neutral molecules. The electron mean free path can be approximated as

$$\lambda_e \approx \frac{k_B T}{\sqrt{2}\,\pi d_m^2 p},$$

where k_B is Boltzmann's constant, T is gas temperature, d_m is the effective molecular collision diameter, and p is gas pressure. Because the mean free path decreases rapidly with increasing pressure, atmospheric-pressure plasma plumes consist of short electron trajectories and dense microdischarge regions. These conditions influence energy transfer efficiency, reactive species production, and the spatial confinement of the plasma plume. Understanding this collisional behavior is therefore important when interpreting plume length, reactive species generation, and plasma–tissue coupling in the PMRI system.

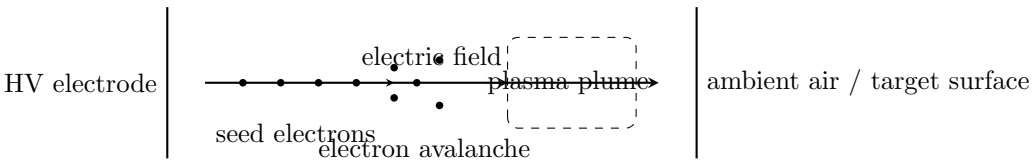


Figure 6: Conceptual illustration of electron avalanche formation in an atmospheric-pressure discharge. A strong electric field accelerates seed electrons, producing collisions and ionization events that generate additional electrons and reactive species. The resulting avalanche process supports formation of the localized plasma plume used for PMRI treatment.

6.3 Potential Laboratory Validation Experiments

Because the PMRI concept has not yet been experimentally evaluated, initial laboratory studies will be necessary to verify key elements of the proposed plasma–plant interaction mechanism before field trials are attempted.

Several practical bench-scale experiments could provide early validation of the concept:

1. **Plasma Exposure of Excised Crown Tissue**

   Segments of freshly harvested *Rubus* crown tissue could be exposed to controlled CAP treatment under laboratory conditions. Subsequent analysis could evaluate structural changes in lignin and cell wall polymers using techniques such as FTIR spectroscopy, Raman spectroscopy, or scanning electron microscopy.

2. **Reactive Species Flux Measurement**

   Chemical probes, optical emission spectroscopy, or ozone sensors could be used to quantify reactive oxygen and nitrogen species (ROS/RNS) concentrations in the plasma plume near the tissue surface. These measurements would provide a quantitative estimate of oxidative exposure during treatment.

3. **Plasma Penetration Depth Studies**

   Cross-sections of treated crown tissue could be examined to determine the depth of oxidative modification. Histological staining, fluorescence imaging, or micro-CT scanning could help identify localized structural damage within the outer millimeters of tissue.

4. **Crown Viability Assays**

   Biochemical viability assays, such as tetrazolium chloride staining, could be used to assess metabolic activity in treated versus untreated crown tissues. These assays would help determine whether plasma exposure reduces tissue vitality in regions responsible for bud activation and regrowth.

5. **Controlled Regrowth Experiments**

   Excised crowns or potted *Rubus* plants could be subjected to mechanical cutting followed by plasma treatment in greenhouse conditions. Regrowth rate, cane emergence, and biomass production could then be compared against untreated controls.

Such laboratory experiments would allow controlled investigation of plasma–tissue interactions while eliminating many environmental variables present in field conditions. Results from these studies would provide important empirical guidance for subsequent small-plot field trials of the PMRI concept.

6.4 Surface ROS Dose Mapping

One of the most informative early validation experiments for the PMRI concept is direct measurement of reactive oxygen and nitrogen species (ROS/RNS) delivery at the crown surface under representative operating conditions. Because plasma–plant interactions are ultimately mediated by the local flux of chemically active species, quantifying the oxidative exposure delivered to the treatment surface provides an important first-order validation of the proposed mechanism.

If direct measurement of lignin structural modification or plant physiological response is not yet feasible, demonstration of repeatable ROS delivery at crown-relevant distances (3–10 cm) can provide an essential experimental foundation for subsequent biological studies.

Experimental goal. The objective of this experiment is to demonstrate that the PMRI configuration produces a localized and repeatable oxidative footprint at the target treatment distance and duty cycle used during field operation.

Minimal experimental setup. A simple bench-scale setup can be used to characterize oxidative exposure using inexpensive chemical indicator substrates. Required components include:

- PMRI plasma source or equivalent bench plasma device
- Mechanical stand or clamp to maintain fixed electrode-to-surface distance
- Chemical indicator substrates such as:
  - potassium iodide (KI)–starch paper for ozone and oxidant detection
  - Griess reagent strips as a proxy indicator for reactive nitrogen species
  - Amplex Red reagent films for hydrogen peroxide detection (optional)
- A digital camera or smartphone for consistent image capture under controlled lighting

Experimental procedure. The electrode-to-surface distance is fixed (for example 5 cm) and plasma is applied using the treatment duty cycle described in Section 2.2 (e.g., three 10 s exposures separated by 5 s pauses). A standardized moist surface is used as a surrogate substrate, such as damp filter paper, a thin agar gel plate, or excised crown tissue wrapped in moist gauze.

Chemical indicator substrates are placed at the treatment surface and exposed to the plasma plume. The resulting colorimetric response is recorded and quantified.

Measurements may include:

- oxidant footprint area (cm^2)
- relative color intensity using grayscale image analysis
- repeatability across multiple trials (e.g., $N = 10$)

Additional experiments can evaluate environmental sensitivity by repeating the measurements across different operating conditions, including:

- treatment distance (3, 5, and 10 cm)
- surface humidity (dry versus misted)
- airflow conditions (still air versus moderate airflow)

Expected outcomes. These measurements produce spatial maps of oxidative exposure such as oxidant footprint area as a function of treatment distance, oxidant intensity as a function of duty cycle, and humidity-dependent enhancement of reactive species generation.

Even when expressed as relative measurements, such data provide useful evidence that the plasma plume delivers localized oxidative exposure at the crown surface.

Relevance to the PMRI mechanism. This experiment directly tests the first stage of the proposed interaction chain:

plasma discharge → ROS/RNS delivery at crown surface → tissue oxidation → biological response.

If reproducible ROS delivery cannot be demonstrated at the treatment surface, subsequent biological effects would be unlikely. Conversely, confirmation of a repeatable oxidative footprint at crown-relevant distances provides strong support for the physical feasibility of plasma-assisted crown weakening.

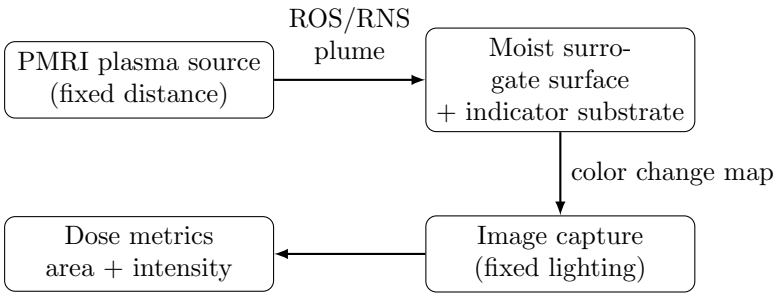


Figure 7: Bench-scale "surface ROS dose mapping" experiment for PMRI validation. A fixed-distance plasma plume is applied to a moist surrogate surface with a chemical indicator substrate; the resulting colorimetric response is imaged under controlled lighting and quantified as footprint area and intensity.

Because this experiment requires only a small laboratory setup and inexpensive indicator materials, it represents a practical early validation step for the PMRI concept prior to more complex biochemical or field studies.

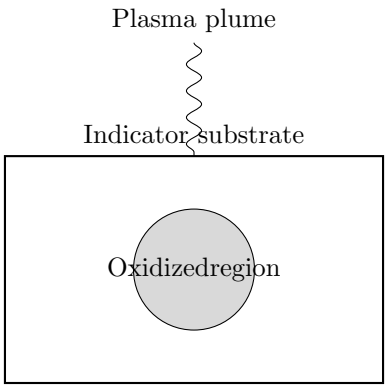


Figure 8: Conceptual visualization of ROS indicator response during the surface dose mapping experiment. Plasma exposure produces a localized oxidative footprint on the chemical indicator substrate, which can be quantified by area and color intensity.

6.5 Future Research Directions

The conceptual framework presented in this paper suggests several promising avenues for future investigation. Initial laboratory experiments could focus on controlled plasma exposure of excised Rubus crown tissue to quantify structural modification of lignin and cell wall polymers using spectroscopic techniques such as FTIR, Raman spectroscopy, or scanning electron microscopy.

Subsequent greenhouse or small-plot field trials could evaluate regrowth suppression following plasma treatment relative to conventional mechanical cutting controls. Measurements of crown viability, carbohydrate depletion, and microbial colonization would help clarify the biological pathways linking plasma exposure to reduced regenerative capacity.

Further engineering development may explore optimization of plasma operating parameters, including voltage amplitude, exposure duration, and treatment distance. Portable plasma devices with improved energy efficiency and automated duty-cycle control could increase treatment throughput and operational safety.

Finally, integration of plasma treatment tools with emerging robotic or drone-assisted vegetation management systems may enable scalable deployment of targeted crown treatments across large restoration landscapes.

7 Conclusion and Outlook

The Pulsed Micro-Plasma Root Interface (PMRI) introduces a novel non-thermal strategy for managing woody perennial invasive plants through targeted oxidative weakening of root crown tissues. By applying cold atmospheric plasma directly to exposed *Rubus* crown surfaces following mechanical cutting, the proposed method leverages well-established plasma chemistry to induce localized oxidative damage in lignified vascular tissues. Cleavage of lignin β-O-4 aryl ether linkages, disruption of cellular membranes, and increased susceptibility to microbial colonization together provide a plausible multi-stage pathway for reducing crown viability and suppressing regrowth.

Compared with conventional control techniques—including mechanical excavation, repeated mowing, herbicide application, and flame weeding—the PMRI concept offers several potential advantages:

- Non-thermal operation that avoids ignition risks associated with flame-based methods
- Absence of persistent chemical residues in soil or water
- Minimal soil disturbance during treatment
- Targeted application to individual crowns following mechanical cutting

These characteristics make the method particularly attractive for managing invasive *Rubus* populations on steep slopes, terrace edges, and environmentally sensitive landscapes where conventional methods pose erosion or contamination risks.

A further advantage of the approach is its practical accessibility. A functional prototype device can be constructed from inexpensive, widely available components, placing experimental validation within reach of small research teams, land managers, and community restoration groups. Because the treatment protocol integrates naturally with existing mechanical cutting practices, initial field trials could be conducted rapidly using small-plot randomized designs within a single growing season.

Several key research questions remain and should be addressed through controlled experimental studies:

- Quantification of plasma penetration depth in woody crown tissues
- Measurement of structural modification in lignin and cell wall polymers following treatment
- Determination of dose–response relationships between plasma exposure and crown vitality
- Evaluation of long-term regrowth suppression under realistic field conditions
- Assessment of interactions between plasma treatment and soil microbial communities

Beyond invasive plant management, the concept highlights a broader opportunity for applying atmospheric plasma technologies to targeted plant tissue modification in ecological and agricultural contexts. Portable plasma tools may enable new forms of selective, residue-free vegetation management that bridge advances in plasma chemistry with practical land stewardship.

The Pulsed Micro-Plasma Root Interface therefore represents a promising conceptual approach that warrants experimental investigation. With modest prototype development and collaborative field trials, this concept could evolve from a theoretical proposal into a practical tool for environmentally sensitive invasive species control.

Future work will focus on experimental validation of plasma–plant interactions in woody crown tissues, quantitative characterization of reactive species exposure during field treatment, and development of optimized portable plasma devices for ecological vegetation management. These studies will determine whether targeted plasma-assisted crown weakening can become a practical tool within integrated invasive species management strategies.

The PMRI concept illustrates how advances in atmospheric plasma technology may enable a new class of targeted, low-cost, residue-free vegetation management tools. If experimentally validated, plasma-assisted crown weakening could become a practical component of integrated invasive plant control strategies, particularly in environmentally sensitive or difficult-to-access landscapes.

8 Acknowledgements

The author gratefully acknowledges the growing body of research in plasma agriculture and plasma medicine that has provided the scientific foundation for this work. In particular, studies on cold atmospheric plasma interactions with lignocellulosic materials and biological tissues have informed the conceptual development of the Pulsed Micro-Plasma Root Interface (PMRI).

No external funding was received for this research. The work was conducted independently as part of an ongoing exploration of low-cost technological approaches for ecological restoration and invasive species management.

The author also acknowledges the open scientific literature and publicly accessible technical resources that enabled the synthesis of plasma chemistry, plant physiology, and invasive plant management concepts presented in this study.

A Prototype Hardware Bill of Materials

Table 5 lists example components suitable for constructing a prototype Pulsed Micro-Plasma Root Interface (PMRI) device using commercially available hardware. The parts listed represent typical low-cost components commonly used in experimental atmospheric plasma devices.

Table 5: Example bill of materials for a prototype PMRI device.

Component	Example Part	Cost (USD)	Notes
ZVS driver module	12–24V zero-voltage switching driver	10–15	High-frequency driver for flyback transformer
Flyback transformer	Compact HV flyback module	15–30	Generates high voltage for plasma discharge
Quartz capillary tube	3–4 mm inner diameter	5–10	Dielectric barrier and plasma channel
Needle electrode	Stainless steel rod	¡1	Inner electrode for discharge
Outer electrode	Brass or copper ring	2–5	Stabilizes plasma plume
Battery pack	12V LiPo battery (3–5Ah)	25–40	Portable power supply
Fiberglass pole	1.5–2 m insulated pole	15–30	Provides operator distance and insulation
Miscellaneous wiring	High-voltage silicone wire	5–10	Electrical connections

Total estimated hardware cost for a prototype device is typically under $150 depending on component sourcing.

B Conceptual Electrical Architecture

The PMRI device is based on a compact dielectric barrier discharge (DBD) plasma configuration. In this design, a high-frequency high-voltage driver energizes an electrode contained within a dielectric tube, producing a localized plasma plume at the outlet.

The conceptual electrical architecture consists of the following components:

- A low-voltage DC power source (typically 12 V battery)
- A zero-voltage-switching (ZVS) driver circuit
- A step-up transformer or flyback transformer generating kilovolt-range output
- A dielectric barrier discharge electrode assembly

The simplified system architecture is:

$$\text{Battery} \rightarrow \text{ZVS Driver} \rightarrow \text{Flyback Transformer} \rightarrow \text{Plasma Electrode}$$

Optional pulse modulation can be implemented using a microcontroller-based gating circuit to produce intermittent plasma bursts during treatment.

C Plasma Chemistry in Humid Air

Atmospheric-pressure plasma operating in ambient air produces a mixture of reactive oxygen and nitrogen species (ROS/RNS) through electron-driven reactions involving oxygen, nitrogen, and water vapor.

Common reactions include:

$$e^- + H_2O \rightarrow OH^\bullet + H^\bullet \tag{1}$$
$$O_2 + e^- \rightarrow 2O^\bullet \tag{2}$$
$$O^\bullet + O_2 \rightarrow O_3 \tag{3}$$
$$N_2 + O^\bullet \rightarrow NO + N^\bullet \tag{4}$$

These reactions produce chemically active species including:

- Hydroxyl radicals ($OH^\bullet$)
- Ozone (O_3)
- Hydrogen peroxide (H_2O_2)
- Nitric oxide (NO)
- Nitrogen dioxide (NO_2)

Hydroxyl radicals are particularly important in humid environments because they are among the most powerful oxidizing agents known and can rapidly modify lignin and other organic polymers.

D Rubus Crown Structure and Regeneration

The crown of *Rubus* plants forms the central regenerative structure connecting above-ground canes to the root system. This crown typically contains:

- Dormant buds capable of producing new canes
- Cambial tissues responsible for vascular growth
- Lignified vascular bundles transporting water and nutrients
- Stored carbohydrate reserves used to fuel regrowth

Following mechanical cutting or disturbance, dormant buds located near the crown surface are activated and initiate rapid shoot formation. These buds rely on stored carbohydrate reserves contained within nearby vascular tissues. Consequently, damage to the outer crown layers can significantly impair the plant's ability to regenerate new canes even when deeper roots remain intact.

E Proposed Field Trial Protocol

Future validation of the PMRI concept could be performed through small-plot randomized field experiments.

A possible experimental design includes:

- Three treatment groups: PMRI, cut-only control, and flame weeding comparison
- Thirty individual crowns per treatment group
- Randomized block layout to reduce environmental bias

Measurements could include:

- Regrowth biomass at 30, 60, and 90 days
- Crown viability using tetrazolium chloride staining
- Lignin structural modification using FTIR spectroscopy
- Soil microbial community shifts using 16S sequencing

Such experiments would provide the first empirical assessment of plasma-assisted crown weakening in woody invasive plants.

F Estimated Treatment Throughput

Treatment throughput depends on exposure duration and crown density.

Using the treatment protocol described in this paper:

- Plasma exposure per crown: approximately 45 seconds
- Operator repositioning time: approximately 10–15 seconds

This suggests a typical treatment rate of:

- 50–70 crowns per hour per operator
- 400–600 crowns per day assuming 8 hours of operation

Actual field performance will depend on vegetation density, terrain, and operator experience.

G Typical Plasma Parameter Ranges

Table 6 summarizes representative operating parameters for small atmospheric plasma devices comparable to the PMRI concept.

Table 6: Representative plasma operating parameters for handheld atmospheric plasma devices.

Parameter	Typical Range
Voltage amplitude	8–20 kV
Operating frequency	10–40 kHz
Electrical power	5–20 W
Plasma plume length	1–3 cm
Gas temperature	¡ 40 °C
Treatment distance	3–10 cm

H Conceptual Plasma–Plant Interaction Model

The Pulsed Micro-Plasma Root Interface (PMRI) concept relies on a chain of interacting physical, chemical, and biological processes that connect plasma generation to eventual suppression of plant regrowth. Although the system has not yet been empirically tested, the proposed mechanism integrates well-established principles from atmospheric plasma physics, reactive oxygen chemistry, lignocellulosic material science, and plant physiology.

The conceptual model can be described as a sequential cascade of processes:

1. Plasma generation in ambient air produces energetic electrons and ionized species.
2. Electron-driven reactions generate reactive oxygen and nitrogen species (ROS/RNS).
3. These reactive species oxidatively modify lignin and other structural polymers in crown tissues.
4. Structural and metabolic damage disrupts crown physiology and stored energy mobilization.
5. Reduced crown vitality results in weaker regrowth and progressive exhaustion of the plant.

H.1 Mechanistic Interaction Layers

The interaction between plasma treatment and plant physiology can be interpreted across four conceptual layers.

Table 7: Conceptual layers linking plasma physics to plant regrowth suppression.

Layer	Primary Processes	Expected Biological Effect
Plasma Physics	Electron impact reactions, ionization, dielectric barrier discharge	Generation of reactive plasma environment
Plasma Chemistry	Formation of ROS/RNS (OH, O_3, H_2O_2, NO_x)	Oxidative chemical agents interact with tissues
Plant Tissue Interaction	Lignin oxidation, membrane damage, increased porosity	Structural weakening of crown tissues
Plant Physiology	Reduced carbohydrate transport and bud activation	Suppression of regrowth and crown vitality

H.2 Conceptual Interaction Diagram

Figure 9 illustrates the conceptual interaction pathway linking plasma generation to suppression of regrowth in *Rubus* crowns.

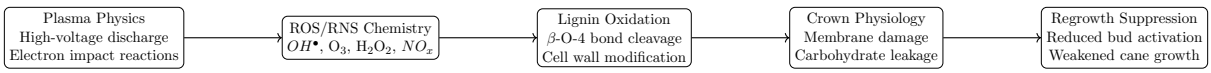


Figure 9: Conceptual model linking atmospheric plasma generation to suppression of *Rubus* crown regrowth through reactive species chemistry and plant tissue modification.

This framework highlights the interdisciplinary nature of the PMRI concept. Plasma physics determines the generation of reactive species, plasma chemistry governs the formation and lifetime of ROS/RNS, and plant physiology ultimately determines how oxidative damage translates into reduced regenerative capacity.

Future experimental studies can test individual stages of this model by measuring reactive species concentrations near the crown surface, characterizing lignin structural changes in treated tissues, and quantifying regrowth suppression following plasma exposure.

I Plasma Energy and Reactive Species Dose Estimate

Although no direct measurements were conducted in this study, it is useful to estimate the approximate energetic scale of plasma treatment applied to a *Rubus* crown. Such estimates provide an order-of-magnitude assessment of whether the proposed treatment could plausibly induce measurable chemical modification of lignified tissues.

I.1 Electrical Energy Delivered

The prototype PMRI device described in this paper is expected to operate in the range of 5–20 W of electrical power during plasma generation. Using the treatment protocol described in Section 2 (approximately 45 s exposure per crown), the total electrical energy delivered per treated crown can be approximated as:

$$E = P \times t \tag{5}$$

where P is electrical power and t is exposure time.

Assuming a nominal operating power of 10 W:

$$E \approx 10\,\mathrm{W} \times 45\,\mathrm{s} = 450\,\mathrm{J} \tag{6}$$

Thus each treatment event delivers on the order of several hundred joules of electrical energy to the plasma discharge.

I.2 Reactive Species Production

In atmospheric plasma systems, only a fraction of the electrical energy is converted into chemically active species. Nevertheless, even modest conversion efficiencies can generate substantial quantities of reactive molecules.

Reactive oxygen species produced in humid air plasmas typically include:

- Hydroxyl radicals ($OH^{\bullet}$)
- Atomic oxygen ($O^{\bullet}$)
- Ozone (O_3)
- Hydrogen peroxide (H_2O_2)
- Nitric oxide (NO) and nitrogen oxides

Laboratory plasma reactors operating at similar power levels have demonstrated production of reactive species at concentrations sufficient to induce measurable oxidation of lignocellulosic materials.

I.3 Estimated Treatment Area

The plasma plume produced by handheld dielectric barrier discharge devices typically extends 1–3 cm from the electrode tip. If the effective treatment diameter is approximated as 2 cm, the exposed surface area per treatment is:

$$A \approx \pi r^2 \tag{7}$$

$$A \approx \pi(1\,\mathrm{cm})^2 \approx 3.1\,\mathrm{cm}^2 \tag{8}$$

This corresponds to a localized treatment region focused on the exposed crown surface.

I.4 Implications for Lignin Modification

Lignin is susceptible to oxidative cleavage of β-O-4 aryl ether bonds when exposed to strong oxidizing agents such as hydroxyl radicals and ozone. Plasma-assisted pretreatment methods used in lignocellulosic biomass processing have demonstrated that relatively modest energy inputs can produce detectable structural modification of lignin networks.

Given the estimated energy delivery and localized treatment area, it is plausible that PMRI treatment could produce measurable oxidative modification within the outer few millimeters of crown tissue. Even partial degradation of lignified vascular structures may be sufficient to disrupt carbohydrate transport and reduce regenerative capacity.

I.5 Limitations of the Estimate

These calculations represent simplified order-of-magnitude estimates and should not be interpreted as direct measurements. Actual treatment efficacy will depend on many factors including plasma efficiency, crown moisture content, tissue composition, and environmental conditions.

Future experimental studies should directly measure:

- plasma power coupling efficiency
- reactive species concentrations near the crown surface
- depth of lignin modification within treated tissues
- regrowth suppression following treatment

Such measurements would allow the conceptual energy model presented here to be refined and validated.

J Comparison with Existing Invasive Control Methods

Management of woody invasive plants such as *Rubus* species typically relies on a combination of mechanical, chemical, and thermal approaches. Each method has distinct advantages and limitations depending on terrain, environmental sensitivity, and available labor.

Table 8 compares the conceptual Pulsed Micro-Plasma Root Interface (PMRI) approach with several commonly used invasive plant control techniques.

In many restoration and land management contexts, no single control method provides complete eradication of established woody invasives. Instead, integrated management strategies are commonly employed, combining mechanical cutting with chemical or thermal treatments.

Within such integrated approaches, the conceptual PMRI system could potentially serve as a targeted crown-weakening step following mechanical cutting. By oxidatively damaging crown tissues without disturbing surrounding soil, the method may offer particular advantages

Table 8: Comparison of PMRI with commonly used invasive plant control methods.

Method	**Primary mechanism**	**Soil disturbance**	**Chemical residues**	**Typical limitations**
PMRI (conceptual method)	Localized plasma-generated ROS/RNS oxidize lignified crown tissues following cutting, aiming to weaken cambial/vascular function and promote decline	Minimal	None	Limited penetration depth; requires prototype validation and controlled field trials; likely needs follow-up treatments
Herbicide application	Systemic chemical disruption of plant metabolism and translocation to crowns/roots	Minimal	Present	Environmental concerns and regulatory restrictions; potential non-target impacts; public acceptance constraints
Flame weeding	Thermal destruction of aboveground tissues; limited heating of lignified crown/root tissues	Minimal	None	Fire risk; reduced effectiveness on established woody crowns; weather constraints (wind, drought bans, rain)
Mechanical digging	Physical removal of crowns and root structures	High	None	Soil disturbance and erosion risk on slopes; high labour and equipment burden; may be impractical on terrace edges
Cut-only management	Repeated cutting to progressively exhaust plant energy reserves	Minimal	None	Typically requires multiple seasons of repeated effort; regrowth can be rapid from crown buds and stored carbohydrates

in environments where erosion control, riparian protection, or chemical-free management are priorities.

Because the PMRI approach has not yet been experimentally validated, the comparison presented here should be interpreted as conceptual. Future field trials will be required to determine the practical effectiveness, treatment efficiency, and operational feasibility of plasma-assisted invasive plant control relative to established methods.

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